# FABIAN ALVAREZ-PRIMO

**OPERATIONS SYSTEMS CONSULTANT | AI & AUTOMATION ENGINEER | GOOGLE SHEETS · PYTHON · GCP · BIOTECH DOMAIN EXPERT**

Savannah, GA · +1 (915) 603-2117 · fpalvarez23@gmail.com · github.com/fabzy4L

## About

I build the tool when the tool doesn't exist. At Amazon, that meant a three-layer Ship Dock intelligence system — browser automation scraping live WIP data, a real-time Google Sheets analytics layer, and a v3.0 staffing framework — because native tooling had a gap and 400-unit dock rates don't wait. Stack I reach for: Google Sheets and Apps Script for structured ops data, Python for pipelines and PDF parsing, GCP for serverless orchestration, Gemini and OpenAI APIs for synthesis. PhD in biomedical engineering — matters when the domain is genuinely complex or the data is weird. Full portfolio: github.com/fabzy4L

## Skills & Tools

| | |
|---|---|
| Google Sheets (Advanced) | Google Apps Script |
| Python (pandas, numpy, PyMuPDF) | R (Tidyverse, Shiny) |
| SQL | ChromaDB / Vector Stores |
| Google Cloud Platform (Functions, Run, Storage, TTS) | Streamlit |
| Claude & Gemini API Integration | RAG Pipeline Architecture |
| KPI Dashboard Development | Workflow Automation |
| Data Pipeline Design | Process Documentation |
| Lean / Six Sigma | CLIA/GMP Compliance Systems |
| Lab Operations Automation | Biotech Domain Intelligence |

## Core Competencies

| | | |
|---|---|---|
| Automation & Data Analytics (Python, R, SQL) | Regulated Operations (CLIA, COLA, GMP-adjacent) | Quality Systems & Compliance |
| KPI Development & Performance Monitoring | Team Leadership & Development (60+ Direct Reports) | Cross-functional Collaboration |
| Scientific & Technical Depth (PhD, 8 Publications) | Biomanufacturing Scale-up & cGMP Processes | Process Optimization & Lean Manufacturing |

## Professional Experience

### Operations Manager II | Amazon | Savannah, GA | 2024 – Present

- Manage operations team of **60+ associates** across a compressed 4-day schedule in a high-volume fulfillment environment, maintaining **99.7% quality compliance**.
- Implemented Python-based labor forecasting models, reducing scheduling variance and optimizing resource allocation across shifts.
- Led cross-functional process improvement initiatives applying Lean methodologies, driving measurable efficiency gains and waste reduction.
- Established quality control protocols and real-time KPI dashboards for performance monitoring across all operational metrics.
- Developed and delivered team training programs improving onboarding efficiency and accelerating time-to-productivity; trained **15 Process Assistants**, managing up to **9 concurrently**, building a structured PA development pipeline where none previously existed.

- Engineered a three-layer Ship Dock intelligence system — a Tampermonkey browser automation script scraping Amazon's Foresight WIP tool across paginated chute/CPT data, a real-time Google Sheets analytics layer tracking recirculation root causes, staffing ratios, DPMO, and hourly throughput, and a v3.0 staffing and rate management tool — replacing entirely manual workflows with an integrated data pipeline purpose-built around gaps in existing tooling.
- Independently engineered a suite of Google Sheets and Apps Script production tools deployed across active fulfillment operations: Thrive PA review automation (narrative generation from performance data), FORESIGHT volume allocation formula (CART/FLATS split logic across paginated chute data), and RTB Comment Generator (pandas-based UPH analysis with rate-dilution logic for new-hire cohorts) — replacing entirely manual workflows with structured, repeatable data pipelines.
- Drove an **82% improvement in dock rate (oprerational cost efficiency) from 220 → 400** and reduced process recirculation from **5% to below 2%** through data-driven staffing models, chute-level congestion diagnostics, and process standardization across material paths and Vendor Returns operations — the latter built from scratch with no prior operational structure at site.
- Automated associate scheduling and productivity tracking via Thrive, eliminating manual entry bottlenecks and enabling real-time visibility into labor utilization and shift performance across the dock.
- Identified systemic tooling gaps in ship dock operations, independently engineered cross-functional solutions (automation + analytics + staffing frameworks), and drove org-wide adoption — exemplifying the Invent & Simplify arc: gap identified → bridge built → org caught up.

---

## Ventures & Technical Projects

### LabOps.AI | Founder & Technical Lead | 2024 – Present

- AI-native literature intelligence platform for biomanufacturing facilities targeting cell and gene therapy operators.
- Architected end-to-end pipeline: bioRxiv/PubMed ingestion → Unpaywall PDF resolution → PyMuPDF parsing → ChromaDB vector storage → Gemini 2.5 Flash synthesis with structured JSON extraction of Critical Quality Attributes (CQAs) and Critical Process Parameters (CPPs).
- Tech stack: Python 3.13, Gemini 2.5 Flash, ChromaDB, SQLite (WAL mode), PyMuPDF, Unpaywall API.
- Pre-seed stage; literature intelligence layer production-ready.

### biotech-intel | Engineer | 2024 – Present

- Multi-source intelligence pipeline aggregating bioRxiv, PubMed, ClinicalTrials.gov, FDA CDER, and SEC Edgar filings into a unified daily briefing synthesized via GPT-4o-mini.
- Bridges scientific publication tracking with regulatory and financial filings — rare cross-domain signal in biotech competitive intelligence.
- Tech stack: Python, pandas, requests, OpenAI API, lxml.

### BLKPHXLABS AI_BRIEFING | Engineer | 2024 – Present

- Serverless autonomous pipeline: Google Alerts → Gmail → GCP Cloud Function → Gemini script synthesis → MoviePy video production → cloud-hosted dashboard.
- Demonstrates low-cost GCP orchestration pattern (Apps Script as scheduler, GCP as compute) for automated media generation.
- Tech stack: GCP (Functions, Run, Storage, TTS), Google Apps Script, MoviePy, Gemini API.

---

### Post-doctoral Research Fellow | Texas Tech University Health Sciences Center (TTUHSCEP) | El Paso, TX | 2022 – 2023

- Led iPSC reprogramming project directing non-integrating conversion of human PBMCs into clinically relevant induced Pluripotent Stem Cells under **cGMP-compliant** protocols for neuropsychiatric disease modeling.
- Managed multi-experiment research data pipelines applying statistical analysis (R, Python) for transcriptomic and biomarker datasets.
- Oversaw biobanking operations including sample intake, processing, storage, and chain-of-custody documentation to maintain regulatory integrity.

**Clinical Laboratory Scientist & Supervisor | West Texas Pain Institute (WTPI) | El Paso, TX | 2021 – 2022**

- Ensured full **CLIA and COLA regulatory compliance** across all laboratory operations, overseeing **1,300+ patient samples monthly** within a **9-member** team.
- Operated and maintained advanced analytical instrumentation including LC/MS confirmatory systems, Enzyme Immunoassay platforms (CEDIA, DRI, QMS, EIA), and DNA next-generation sequencing.
- Managed LIS bioinformatics system (Stratus DX) and patient EMR records (eClinicalWorks), enforcing documentation accuracy and OSHA compliance.

**Doctoral Research Assistant | Inspired Materials & Stem Cell Tissue Engineering Lab (IMSTEL), UTEP | El Paso, TX | 2017 – 2020**

- Achieved **30% improvement** in 3D printer manufacturing precision through hardware modifications on cardiovascular tissue composite fabrication systems.
- Mentored and trained **30+ undergraduate students** on electron microscopy and 3D bioprinting, developing structured onboarding curricula and hands-on training protocols.
- Managed cross-functional lab operations — coordinating equipment schedules, reagent procurement, SOPs, and data management across **5+ concurrent projects**.

## Education

**Ph.D. in Biomedical Materials Science & Engineering** | The University of Texas at El Paso | 2016 – 2020

**B.S. in Materials Engineering** | The University of Texas at El Paso | 2010 – 2015

## Publications & Technical Portfolio

PhD-level researcher with **8** peer-reviewed publications in biomaterials and tissue engineering. Scientific depth applied to AI pipeline architecture and regulated operations contexts. Full technical portfolio: github.com/fabzy4L

Tau Beta Pi Engineering Honor Society | Alpha Chi National Honor Society | Alpha Sigma Mu Materials Honor Society | Society for Biomaterials (President, 2 terms)